COMPARISON OF THE DIFFERENTIAL FORMS ON THE BLOWUP, THE EXCEPTIONAL DIVISOR AND THE PUNCTURED TOTAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The projection formula. Cohomology, label cohomology-lemma-projection-formula (invoked at proof line 2877). It rewrites p_* of the product decomposition of $\Omega^a_{P/S}$ as a direct sum of tensor products with $\operatorname{pr}_{1,*}\operatorname{pr}_2^*\Omega^s$.
- The computation of the Hodge cohomology of projective space, and the section on projective space. Same chapter derham.tex, Section \ref{section-projective-space} (invoked at proof line 2884) and label lemma-hodge-cohomology-projective-space (invoked at proof line 2886). They show that $\operatorname{pr}_{1,*}\operatornam 1}_S/S}$ vanishes except for $s = 0$, which settles part (2).
- The direct sum decomposition of the pushforward along the punctured line bundle and the exact sequences for its graded pieces. Same chapter derham.tex, Section \ref{section-line-bundle} (invoked at proof line 2893) and labels lemma-push-omega-a (invoked at proof lines 2894 and 2900) and lemma-the-complex-for-L-star (invoked at proof line 2900). They give $\pi_*\Omega^a_{L/S} = \Omega^a_{P/S} \oplus \bigoplus_{k\geq 1}\Omega^a_{L/S,k}$ and the short exact sequences exhibiting $\Omega^a_{L/S,k}$ as an extension of twisted pieces, which is how part (3) is reduced.
- The product decomposition of differential forms and the twisted Hodge cohomology of projective space. Same chapter derham.tex, label lemma-de-rham-complex-product (invoked at proof line 2913) and label lemma-twisted-hodge-cohomology-projective-space (invoked at proof line 2918).

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The first decomposes \$\Omega\^a_{P/S}\$ over \$P = Z \times_S \mathbf{P}\^{n-1}_S\$, the second kills the higher cohomology of the twisted pieces \$\mathcal{O}_Z \boxtimes \Omega\^j(k)\$ for \$k > 0\$, completing part (3).

Source reference. Original lemma 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Navigation pointer for \ref{cohomology-lemma-projection-formula} used inside the counted span (line 2877); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FMA: sources/stacks-derham/pinned-original-source.tex, source lines 871–871. Navigation pointer for \ref{lemma-de-rham-complex-product} used inside the counted span (line 2913); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FMI: sources/stacks-derham/pinned-original-source.tex, source lines 1705–1705. Navigation pointer for \ref{lemma-hodge-cohomology-projective-space} used inside the counted span (line 2886); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUH: sources/stacks-derham/pinned-original-source.tex, source lines 1509–1509. Navigation pointer for \ref{lemma-push-omega-a} used inside the counted span (line 2894); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUF: sources/stacks-derham/pinned-original-source.tex, source lines 1348–1348. Navigation pointer for \ref{lemma-the-complex-for-L-star} used inside the counted span (line 2900); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUK: sources/stacks-derham/pinned-original-source.tex, source lines 1644–1644. Navigation pointer for \ref{lemma-twisted-hodge-cohomology-projective-space} used inside the counted span (line 2918); the argument that consumes it is retained in the span.

Source reference. Original section 0FU8: sources/stacks-derham/pinned-original-source.tex, source lines 1250–1250. Navigation pointer for \ref{section-line-bundle} used inside the counted span (line 2893); the argument that consumes it is retained in the span.

Source reference. Original section 0FMG: sources/stacks-derham/pinned-original-source.tex, source lines 1576–1576. Navigation pointer for \ref{section-projective-space} used inside the counted span (line 2884); the argument that consumes it is retained in the span.

Lemma 0.1. *For $a \geq 0$ we have*

1. *the map $\Omega^a_{X/S} \to b_*\Omega^a_{L/S}$ is an isomorphism,*
2. *the map $\Omega^a_{Z/S} \to p_*\Omega^a_{P/S}$ is an isomorphism, and*
3. *the map $Rb_*\Omega^a_{L/S} \to i_*Rp_*\Omega^a_{P/S}$ is an isomorphism on cohomology sheaves in degree ≥ 1.*

Proof. Let us first prove part (2). Since $P = Z \times_S \mathbf{P}^{n-1}_S$ we see that

$$\Omega^a_{P/S} = \bigoplus\nolimits_{a=r+s} \mathrm{pr}_1^*\Omega^r_{Z/S} \otimes \mathrm{pr}_2^*\Omega^s_{\mathbf{P}^{n-1}_S/S}$$

Recalling that $p = \mathrm{pr}_1$ by the projection formula (Cohomology, Lemma 01E8) we obtain

$$p_*\Omega^a_{P/S} = \bigoplus\nolimits_{a=r+s} \Omega^r_{Z/S} \otimes \mathrm{pr}_{1,*}\mathrm{pr}_2^*\Omega^s_{\mathbf{P}^{n-1}_S/S}$$

By the calculations in Section 0FMG and in particular in the proof of Lemma 0FMI we have $\mathrm{pr}_{1,*}\mathrm{pr}_2^*\Omega^s_{\mathbf{P}^{n-1}_S/S} = 0$ except if $s = 0$ in which case we get $\mathrm{pr}_{1,*}\mathcal{O}_P = \mathcal{O}_Z$. This proves (2).

By the material in Section 0FU8 and in particular Lemma 0FUH we have $\pi_*\Omega^a_{L/S} = \Omega^a_{P/S} \oplus \bigoplus_{k\geq 1} \Omega^a_{L/S,k}$. Since the composition $\pi \circ 0$ in the diagram above is the identity morphism on P to prove part (3) it suffices to show that $\Omega^a_{L/S,k}$ has vanishing higher cohomology for $k > 0$. By Lemmas 0FUF and 0FUH there are short exact sequences

$$0 \to \Omega^a_{P/S} \otimes \mathcal{O}_P(k) \to \Omega^a_{L/S,k} \to \Omega^{a-1}_{P/S} \otimes \mathcal{O}_P(k) \to 0$$

where $\Omega^{a-1}_{P/S} = 0$ if $a = 0$. Since $P = Z \times_S \mathbf{P}^{n-1}_S$ we have

$$\Omega^a_{P/S} = \bigoplus\nolimits_{i+j=a} \Omega^i_{Z/S} \boxtimes \Omega^j_{\mathbf{P}^{n-1}_S/S}$$

by Lemma 0FMA. Since $\Omega^i_{Z/S}$ is free of finite rank we see that it suffices to show that the higher cohomology of $\mathcal{O}_Z \boxtimes \Omega^j_{\mathbf{P}^{n-1}_S/S}(k)$ is zero for $k > 0$. This follows from Lemma 0FUK applied to $P = Z \times_S \mathbf{P}^{n-1}_S = \mathbf{P}^{n-1}_Z$ and the proof of (3) is complete.

We still have to prove (1). If $n = 1$, then we are blowing up an effective Cartier divisor and b is an isomorphism and we have (1). If $n > 1$, then the composition

$$\Gamma(X, \Omega^a_{X/S}) \to \Gamma(L, \Omega^a_{L/S}) \to \Gamma(L \setminus E, \Omega^a_{L/S}) = \Gamma(X \setminus Z, \Omega^a_{X/S})$$

is an isomorphism as $\Omega^a_{X/S}$ is finite free (small detail omitted). Thus the only way (1) can fail is if there are nonzero elements of $\Gamma(L, \Omega^a_{L/S})$ which vanish outside of $E = 0(P)$. Since L is a line bundle over P with zero section $0 : P \to L$, it suffices to show that on a line bundle there are no nonzero sections of a sheaf of differentials which vanish identically outside the zero section. The reader sees this is true either (preferably) by a local calculation or by using that $\Omega_{L/S,k} \subset \Omega_{L^\star/S,k}$ (see references above). $\square$